



BINARY SYSTEMS

RMF Rev 5 Migrator

User manual for Assessment & Authorization teams converting NIST SP 800-53 Rev 4 policy packages to Rev 5.

VERSION	1.1.0
DOCUMENT	User Manual
AUDIENCE	A&A / security assessors

Contents

01	1. What this tool does
02	2. Key concepts
03	3. Before you start
04	4. The workflow
05	5. Baselines
06	6. Dispositions (Rev 4 → Rev 5)
07	7. Document status reference
08	8. Deleting a project (CUI hard delete)
09	9. Exports at a glance
10	10. Interface notes
11	11. Troubleshooting
12	12. Deployment & authentication (summary)
13	13. Version & support

A guide for Assessment & Authorization (A&A) teams operating the tool day to day. For standing it up in AWS, see [DEPLOYMENT.md](#); for working on the code, see [DEVELOPMENT.md](#).

1. What this tool does

The RMF Rev 5 Migrator converts a system's NIST SP 800-53 **Rev 4** security policy documents into **Rev 5**, keeping a human in the loop at every decision.

For each Rev 4 policy document (`.docx`) you upload, the tool:

1. Splits it into sections by heading.
2. Proposes the Rev 4 controls each section addresses (you review and correct this).
3. Applies NIST's official Rev 4 → Rev 5 crosswalk to find the Rev 5 successors.
4. Drafts updated Rev 5 language per section via Amazon Bedrock (you edit and approve).
5. Exports a **structure-preserving** Rev 5 `.docx` plus a full per-control decision log.
6. Reports package coverage against a baseline and flags gaps, including the Rev 5-new **SR** (supply-chain) family that no Rev 4 document carries.

Everything runs inside your own AWS account. Document content and model prompts/responses are never written to logs.

2. Key concepts

TERM	MEANING
Project	One system's A&A package. Holds one or more policy documents and a chosen baseline.
Document	A single uploaded Rev 4 <code>.docx</code> policy.
Baseline	The Rev 5 control set coverage is measured against (FedRAMP level, FIPS 199 level, DoD/CNSSI 1253, or generic 800-53).
Section	A heading-delimited block of a document. Mapping, drafting, and export all operate per section.
Mapping	The set of Rev 4 control IDs a section addresses. LLM-proposed, human-confirmed.
Disposition	What happened to a Rev 4 control in Rev 5 (see §6).
Draft	The proposed Rev 5 language for a section, edited and approved by a reviewer.
Decision log	Per-control audit trail: Rev 4 source, disposition, Rev 5 target, who approved, when.
Coverage	How much of the chosen baseline the approved drafts satisfy.

3. Before you start

- **Access.** In a production (private) deployment the app is reached through your organization's internal portal or signing proxy; your existing network/identity controls gate it. There is no in-app login.
 - **Input format.** Policy documents must be `.docx`. The parser reads paragraphs **and table cells** in reading order, so requirements stated inside tables are captured.
 - **CUI handling.** Uploaded content is encrypted at rest with your customer-managed KMS key and is never logged. Deleting a project (§8) permanently purges every document, export, section, and audit record, including all prior S3 versions.
-

4. The workflow

The interface is a four-step pipeline. The stepper at the top of a document shows where you are: **1 Mapping review** → **2 Rev 5 editor** → **3 Export** → **4 Coverage**.

4.1 Create a project and upload a document

On the landing screen:

1. Under **Projects**, type a project name, pick the **baseline** the system is authorized at, and select **Create**.
2. With the project selected, use the file picker under **Documents** to upload a Rev 4 `.docx`.

Upload registers the document, sends the bytes straight to encrypted storage, and automatically starts parsing. Parsing then chains into control mapping on its own. The status pill advances `upload_pending` → `parsing` → `mapped`; the list refreshes itself while work is in progress. When a document reaches **mapped**, select **Open**.

4.2 Step 1 – Mapping review (the human checkpoint)

This is the gate: nothing is drafted until you approve the mapping.

Each row is one section, showing the section heading, the **proposed Rev 4 controls** (editable), the model's **confidence**, and the mapping **state**.

- Correct any row's control IDs (comma- or space-separated, e.g. `AC-2, AC-6(1)`) and select **Save**.
- When every row is right, select **Approve mapping & continue**.

Approval freezes the control set and automatically starts Rev 5 drafting. After approval the rows become read-only.

4.3 Step 2 – Rev 5 editor

Each section is a card with the original Rev 4 text on the left and the **proposed Rev 5 draft** on the right. The card header shows the Rev 5 target control(s) and the **disposition** (see §6) for each

mapped Rev 4 control.

Per section you can:

- **Edit** the Rev 5 text and select **Save**.
- Expand **Suggestions** for the model's improvement notes.
- Select **Ask assistant** to open a per-section chat — ask it to refine wording or explain a control. The conversation is scoped to that section and its draft.
- Select **Approve section** once the language is final.

The header counter (`n/total approved`) tracks progress. **Every generated draft must be approved** before export unlocks. Editing an approved draft afterward reverts it to unapproved and revokes any prior export.

4.4 Step 3 – Export

Available once every draft is approved (status **review_approved**).

- **Generate Rev 5 .docx** builds a new document that preserves the original's structure — headings, styles, tables, and unmapped boilerplate stay put; only the mapped section bodies are replaced with approved Rev 5 text. This runs in the background; the button shows progress.
- **Download Rev 5 .docx** downloads the generated file once ready.
- **Download decision log (CSV)** downloads the per-control audit trail.

If the document is not ready, a banner tells you to approve the mapping and drafts first.

4.5 Step 4 – Coverage dashboard

Project-level, spanning every document in the package.

- Pick a **baseline** (or use the project default) and **Refresh**.
- The metric shows coverage percentage and covered/total control counts.
- **Baseline gaps** lists required controls no approved draft addresses.
- **New in Rev 5, not covered** lists Rev 5-additions with no Rev 4 predecessor; the **SR supply-chain family is highlighted** because it is the most common blind spot in a Rev 4 → Rev 5 move.
- **Download conversion matrix (CSV)** — one row per Rev 4 control across the package.
- **Download OSCAL (JSON)** — a NIST OSCAL component-definition of the approved drafts, for import into a GRC tool.

5. Baselines

Coverage is measured against the baseline you pick. Available values:

BASELINE	NOTES
<code>generic_800_53</code>	The full 800-53 Rev 5 catalog; no impact-level tailoring.

BASELINE	NOTES
<code>fips199_low</code> / <code>fips199_moderate</code> / <code>fips199_high</code>	FIPS 199 categorization levels.
<code>fedramp_low</code> / <code>fedramp_moderate</code> / <code>fedramp_high</code>	Real FedRAMP baselines (156 / 323 / 410 controls). FedRAMP selects a strict superset of the NIST set at each level.
<code>fedramp_li_saas</code>	FedRAMP Tailored LI-SaaS.
<code>dod_cnssi_1253</code>	DoD / CNSSI 1253.

You can override the baseline per view on the coverage dashboard without changing the project's stored default.

6. Dispositions (Rev 4 → Rev 5)

Each mapped Rev 4 control carries a disposition from NIST's official crosswalk, shown in the editor and the decision log:

DISPOSITION	MEANING
same	Carried into Rev 5 unchanged (same ID).
renamed	Same control, new identifier/title.
moved	Requirement relocated to one other Rev 5 control.
incorporated	Folded into one existing Rev 5 control.
split	Requirement divided across more than one Rev 5 control (e.g. <code>AC-13</code> → <code>AC-2</code> , <code>AU-6</code>).
withdrawn	Withdrawn with no Rev 5 successor (e.g. <code>SC-19</code> , <code>SA-12</code>).

`new` controls (Rev 5 additions with no Rev 4 source, such as the `SR` family) do not originate from a document section; they surface as coverage gaps in §4.5.

7. Document status reference

The status pill on a document reflects where it is in the pipeline:

STATUS	MEANING
<code>upload_pending</code>	Registered; bytes not yet confirmed in storage.
<code>uploaded</code>	Bytes present; parsing not started.
<code>parsing</code> / <code>parsed</code>	Splitting into sections.

STATUS	MEANING
mapping / mapped	Proposing controls / ready for mapping review .
mapping_approved	Mapping confirmed; drafting queued.
drafting / drafted	Generating Rev 5 language / ready for the editor .
review_approved	Every draft approved; ready to export .
exporting / exported	Building the Rev 5 <code>.docx</code> / ready to download .
failed	A step failed. Re-trigger it (re-upload, re-open, or re-run export); the tool records which stage failed.

8. Deleting a project (CUI hard delete)

On the landing screen, with a project selected, **Delete project** permanently removes it. You are prompted to type the project's ID to confirm. This purges every document, export, section, mapping, draft, and audit record, and deletes all stored object versions — there is no recovery. Use it when the engagement is done or when a project must be scrubbed.

9. Exports at a glance

ARTIFACT	WHERE	CONTENTS
Rev 5 <code>.docx</code>	Export	Structure-preserving Rev 5 document.
Decision log CSV	Export	Per-control audit trail for one document.
Conversion matrix CSV	Coverage	One row per Rev 4 control across the whole package.
OSCAL JSON	Coverage	NIST OSCAL component-definition of approved drafts.

CSV exports are formula-injection safe.

10. Interface notes

- **Theme.** The toggle at the top right switches between the dark (default) and light treatments; your choice is remembered. Light is print-friendly.
- **Auto-refresh.** Lists and status views poll on their own while work is in progress; you do not need to reload.
- **Errors** appear as an inline banner on the affected screen, stating what went wrong.

11. Troubleshooting

SYMPTOM	WHAT TO DO
Document stuck at <code>failed</code>	Re-open it, or re-upload. The failed stage is recorded; re-triggering resumes from there.
A section maps to the wrong control	Correct it in Mapping review before approving; you cannot edit the mapping after approval.
Export button disabled	Every draft must be approved first. Check the <code>n/total approved</code> counter in the editor.
Coverage looks too high or too low	Confirm the baseline matches the level the system is authorized at (§5).
Draft text looks generic or empty	Edit it directly, or use Ask assistant . Empty drafts are skipped at export so the original text is preserved.

12. Deployment & authentication (summary)

The tool self-hosts in your AWS account via Terraform. Two modes:

- **private (production, default)**. Lambdas run in your VPC; **every API route requires AWS SigV4 (AWS_IAM)**. Browsers reach the API through your internal portal or signing proxy, inheriting your IAM/identity controls. There is no application-level login to manage.
- **public (dev/demo only)**. An unauthenticated API for local testing. **Never put CUI through it.**

Full deployment steps, VPC-endpoint requirements, and the teardown checklist are in `DEPLOYMENT.md`.

13. Version & support

- **Manual version:** 1.1.0
- **Changelog:** `CHANGELOG.md`
- **Source & issues:** <https://github.com/Redirishman/rmf-rev5-migrator>

The tool assists; it does not certify. A qualified reviewer remains responsible for every mapping, draft, and disposition it produces.